

Toward Scalable Bayesian Uncertainty for Machine Learning with the Ray Tracing Sampler

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Abstract

Bayesian inference gives a neural network calibrated error bars, but the samplers that produce it need the full dataset at every step. Minibatch gradients make each step cheap and add noise, and that noise quietly corrupts the distribution the sampler is meant to reproduce. This project studied the ray tracing sampler of Behroozi [1], a gradient-based MCMC method that carries a unit direction on a sphere instead of an unbounded momentum, and asked whether that change in geometry makes minibatch sampling practical. The summer ran in three stages. First, exact full-batch chains on small MNIST networks ($D = 50,890$ and $D = 12,810$) established ground truth and showed what the prior costs at initialization. Second, controlled noise experiments measured how far each sampler can be pushed before it breaks: the critical noise level scales as h^{-1} for ray tracing and $h^{-1/2}$ for Hamiltonian Monte Carlo. Third, the sampler was applied to a real survey problem, regressing $[\alpha/M]$ from Gaia XP spectra for 126,156 giant stars with an 11,394 parameter network, where a minibatch chain produced held-out predictions that are 27% more probable on average than the single best fit, with the gain sitting in the error bars rather than the point predictions. A companion theoretical analysis by the project mentor explains the measured scalings and predicts the published noise tolerances to within 30% with no fitted quantities. What is validated here is robustness to gradient noise. Width honesty, anisotropic noise, and cost at modern network scales remain open.

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🔗 **Code:** <https://github.com/YasirAlsugair/ray-tracing-sampler-experiments>

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Contents

1 Introduction 2

2 Methodology 2

2.1 The sampler 3

2.2 Ground truth on MNIST 3

2.3 Controlled noise experiments 3

2.4 The Gaia campaign 4

2.5 Diagnostics 4

3 Results 4

3.1 What ground truth cost 4

3.2 How much noise each sampler tolerates 5

3.3 The Gaia campaign 5

4 Why the tolerance differs: the geometry of the auxiliary variable 9

5 Discussion 10

6 Limitations and future work 11

7 Conclusion 12

References 13

1 Introduction

A trained neural network returns one number per input. A Bayesian treatment returns a distribution over the weights, and from it a distribution over predictions, which is what lets a model say how uncertain it is. The obstacle is cost. Markov chain Monte Carlo methods that sample the posterior of a network need the gradient of the log-posterior, and computing it exactly means touching every training example at every step. On a survey with 10^5 stars or a dataset with 10^6 images, that is not a practical loop.

The standard fix is a minibatch: estimate the gradient from a small random subset. The estimate is unbiased and it is N/B times cheaper, but it is noisy, and the noise does not average out of a sampler the way it averages out of an optimizer. In stochastic gradient descent the noise is annealed away by a decaying learning rate. In a sampler there is no annealing, because the chain is supposed to keep moving; the noise enters the dynamics at every step and shifts the distribution that the chain settles into. The size of that shift is the central quantity of this project.

Hamiltonian Monte Carlo carries an auxiliary momentum $p \in \mathbb{R}^D$ and moves the state along a trajectory of constant energy. A noisy gradient enters the momentum update, and because the momentum is unbounded, repeated noisy kicks raise its expected magnitude. The chain heats up. The ray tracing sampler of Behroozi [1] replaces the momentum with a unit direction u on the sphere S^{D-1} and propagates a constant-speed ray through a medium of refractive index $n(x) = \pi(x)^{1/(D-1)}$, where Snell's law bends the ray toward higher probability. The noisy kick is projected onto the tangent plane of the sphere, so noise can rotate the direction but cannot change its norm. Heating a momentum makes it hotter without bound. Heating a direction only makes it point somewhere else.

That is a claim about geometry, and it is testable. This report covers the work done to test it over the KAUST Academy Summer School 2026, hosted at the University of Toronto Data Sciences Institute. The project ran in four stages.

- 1. Ground truth first.** Small MNIST networks where full-batch exact chains are feasible, so that every later shortcut has something to be checked against. This produced the $T = 1$ posterior target, the convergence protocol, and the observation that where a chain starts decides what its budget buys.
- 2. Measurement.** Controlled experiments on the noise tolerance of both samplers, on idealized isotropic noise and on real minibatch noise from a Bayesian linear regression, together with a cost frontier in dataset passes per effective sample.
- 3. A real target.** The Gaia XP to $[\alpha/M]$ regression behind Laroche and Speagle [2], run as a full campaign with a noise-model ladder, calibration diagnostics, baselines on the same target, and an exactness certificate from a full-batch finisher.
- 4. Theory.** A companion analysis by the project mentor that explains why the tolerance differs, and whose quantitative predictions are confronted with both the published numbers and the measurements here.

The order matters. Each stage exists because the previous one raised a question that could not be settled inside it.

2 Methodology

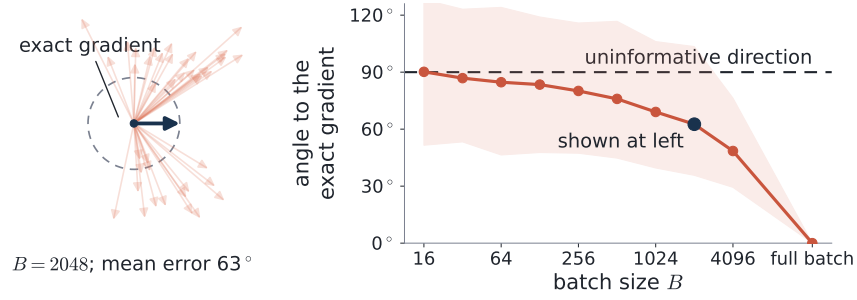


Figure 1 Angle between the minibatch gradient and the exact gradient on the superconductivity Bayesian linear regression. At batch 2,048 the average angle is 63 degrees. The estimate is unbiased, but any single step is pointed substantially wrong.

2.1 The sampler

Both samplers target a posterior $\pi(\theta)$ by augmenting the state with an auxiliary variable and alternating a deterministic move with a refresh of that variable. The difference is where the auxiliary variable lives.

Hamiltonian Monte Carlo draws a momentum $p \sim \mathcal{N}(0, I)$ in $\mathbb{R}^D$ and integrates Hamilton’s equations with leapfrog steps of size h , using $\nabla \log \pi(\theta)$ in the momentum update. Ray tracing draws a direction u uniformly on S^{D-1} and advances the state at unit speed, bending u at each step by a kick built from $\nabla \log \pi(\theta)$ and then renormalizing. Because the step is a geometric distance and the speed is fixed, the gradient can only steer the ray. The refractive index $n(\theta) = \pi(\theta)^{1/(D-1)}$ is chosen so that conserving the basic radiance of the ray is the same thing as sampling in proportion to π . Both samplers accept or reject the endpoint with a Metropolis test, so both are exact at the level of the chain when the gradient is exact.

Under a minibatch gradient neither test is exact, because the log-density itself is estimated. For the minibatch runs here the noise-softened gate of Behroozi [1] was used, which widens the acceptance threshold by an amount set by the measured noise scale σ_{sto} . The gate is a time tax rather than a bias: gated and ungated chains agree on the same target, the gated one just takes longer to get there.

2.2 Ground truth on MNIST

The first experiments used two small MNIST networks: an MLP with $D = 50,890$ parameters and a CNN with $D = 12,810$. Both are small enough that a full-batch exact chain is affordable, which is the point. A $T = 1$ posterior target was derived so that the temperature is not a free knob, and chains were run at full batch with an exact Metropolis test as the reference against which every minibatch shortcut is judged.

2.3 Controlled noise experiments

To measure noise tolerance cleanly, isotropic Gaussian noise of known variance σ^2 was added to exact gradients on Gaussian targets in $D = 16$ to 1024, and the critical noise level σ_c at which each sampler fails its accuracy gate was located as a function of step size h . The same measurement was then repeated with real minibatch noise, on a Bayesian linear regression for the UCI superconductivity dataset ($D = 81, N = 17,010$), where the exact gradient is available for comparison at every step. Figure 1 shows why this matters: at batch 2,048 on that problem the minibatch gradient points 63 degrees away from the exact gradient on average, where 90 degrees would be a random direction.

2.4 The Gaia campaign

The application target is the regression behind Laroche and Speagle [2]: predict $[\alpha/M]$ from Gaia XP coefficients, 55 BP and 55 RP per star [3]. The pristine cut was reproduced at 126,156 giants with APOGEE labels, against 123,804 in the paper, a 1.9% difference. The split used seed 2003, 80/20, giving 100,924 training and 25,232 test stars. The network is a 110-64-64-1 tanh MLP, $D = 11,329$ growing to 11,394 and 11,395 as the noise model gains parameters. Sampling used the vendor minibatch ray tracer with the Eq. 33 gate. The campaign cost roughly \$85 to \$90 of cloud GPU time.

The noise model was built as a ladder, one rung at a time, because it was not clear at the start which part of the problem was mis-specified. Rung 0 uses the catalogue label errors only. Rung 1 adds a global scatter term τ . Rung 1b adds a fitted global scale s . Rung 2 replaces it with a heteroscedastic $\sigma(x)$ predicted by the network. Rung 3 replaces the Gaussian likelihood with a Student- t whose degrees of freedom ν are sampled.

2.5 Diagnostics

Every claim in Section 3 rests on the same three checks. Convergence is judged by drift series (running estimates of weighted norm, log-posterior, and σ_{sto}) read together with the acceptance rate, never separately. Calibration is judged by the standardized residual $z = (y - \hat{y})/\hat{\sigma}$ on held-out stars: its standard deviation should be 1, and it should be 1 within every bin of label error, not just on average. Exactness is judged by restarting from the converged endpoint with an exact full-batch chain and a step-size ladder; if the endpoint is in the right place, the exact chain accepts and stays.

3 Results

3.1 What ground truth cost

The full-batch exact chains on MNIST ran at 97 to 99% acceptance and still never passed stationarity checks at this scale. The exact test is not the bottleneck; the number of steps is. The noise-softened minibatch gate converged on both architectures, at 5.6M minibatch steps for the MLP and 14.6M for the CNN.

The more useful finding was about starting points. Chains launched from low-norm states, the Adam solution or a random initialization, spend most of their budget marching outward toward the shell at $\|\theta\|^2 \sim D$ where the posterior mass actually sits (Figure 2). A prior-aware MAP-transit start, suggested by the project mentor, pre-pays that burn-in: chains launched from transit states accept 92 to 96% at the full tuned step size, while raw bad-fit starts needed step sizes $12\times$ to $350\times$ smaller to accept at all. Where the chain starts decides what its budget buys.

The converged CNN posterior reaches accuracy 0.9892, NLL 0.0854, and ECE 0.0033. A ten-seed deep ensemble reference [4] reaches 0.9904 accuracy and NLL 0.0310. The sampled posterior essentially matches ensemble accuracy, and an NLL gap remains. At a matched review budget of 11% of the test set, the converged chain catches 90% of its own errors at $75\times$ enrichment, the point estimate catches 86% at $52\times$, and the ensemble catches 92% at $93\times$. The flagged sets only half overlap, with Jaccard between 0.32 and 0.52, so these methods are not finding the same hard examples. On corrupted images, per-sample disagreement was the most consistent detector, and confidence and disagreement decouple.

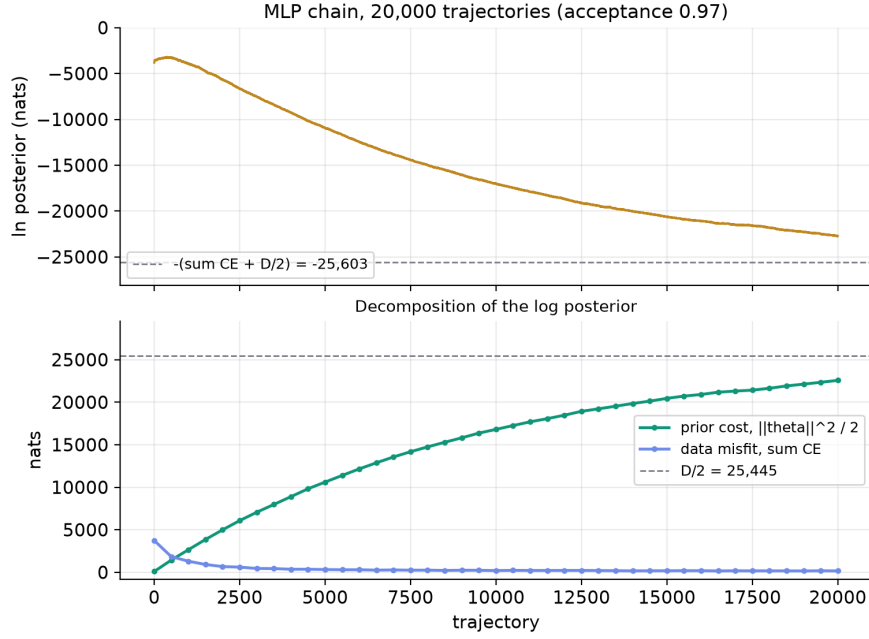


Figure 2 MNIST chains marching from low-norm starting states out toward the shell at $\|\theta\|^2 \sim D$. Most of an untuned chain’s budget is spent on this transit rather than on sampling.

3.2 How much noise each sampler tolerates

Under idealized isotropic gradient noise in $D = 16$ to 1024 , the critical noise level scales as $\sigma_c \propto h^{-1}$ for ray tracing and $\sigma_c \propto h^{-1/2}$ for HMC, and the measured slopes match those exponents (Figure 3, panel a). The practical reading is that halving the step size buys ray tracing twice the noise tolerance and buys HMC only a factor of $\sqrt{2}$.

Under real minibatch noise the picture degrades in a specific way. On the superconductivity regression the local step-size exponent for ray tracing erodes from about 1 toward about 0.6 (panel b). The advantage survives, but the clean scaling does not. Section 4 explains this: real minibatch noise is anisotropic, and anisotropy enters the bias at first order in h , where the isotropic part enters at second order.

The cost frontier (panel c) measures dataset passes per effective sample against batch size. Minibatch HMC needs roughly 30 passes per effective sample and gets worse as the batch shrinks. Minibatch ray tracing sits below the full-batch reference line at small batches.

For scale, Behroozi [1] reports $\sigma_c^2 \approx 3.2$ for ray tracing against 0.012 for HMC at $D = 10,000$, a variance ratio above 250.

3.3 The Gaia campaign

The noise model, not the sampler.. The headline result of the campaign is Figure 4. The standard deviation of the held-out z residual falls from 6.88 using catalogue errors alone, to 6.23 adding a global scatter term, to 0.95 adding a fitted global scale, to 1.08 with a heteroscedastic $\sigma(x)$. A sampler cannot fix a likelihood that is wrong by a factor of six. Most of the work in applying a Bayesian method to a real survey was in the noise model.

The heteroscedastic rung ($D = 11,394$) is the chain of record. Its z standard deviation is 1.08, and by bin of label error it reads 0.97/1.07/1.07/1.07, flat, where the global-scale rung ranged from 0.39 to 1.43. RMSE is 0.0494 and the count of $|z| > 4$ tails is 108.

The Student- t rung ($D = 11,395$) was certified on 2026-08-07 at 8.5M steps with all drift checks

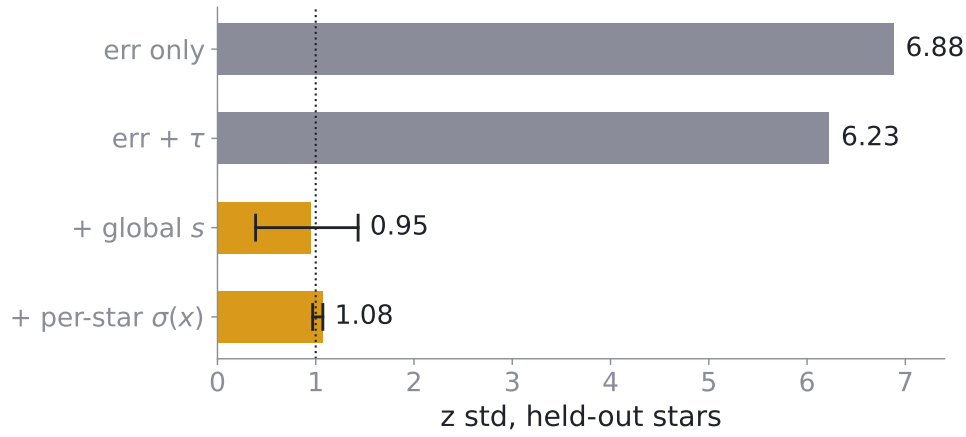
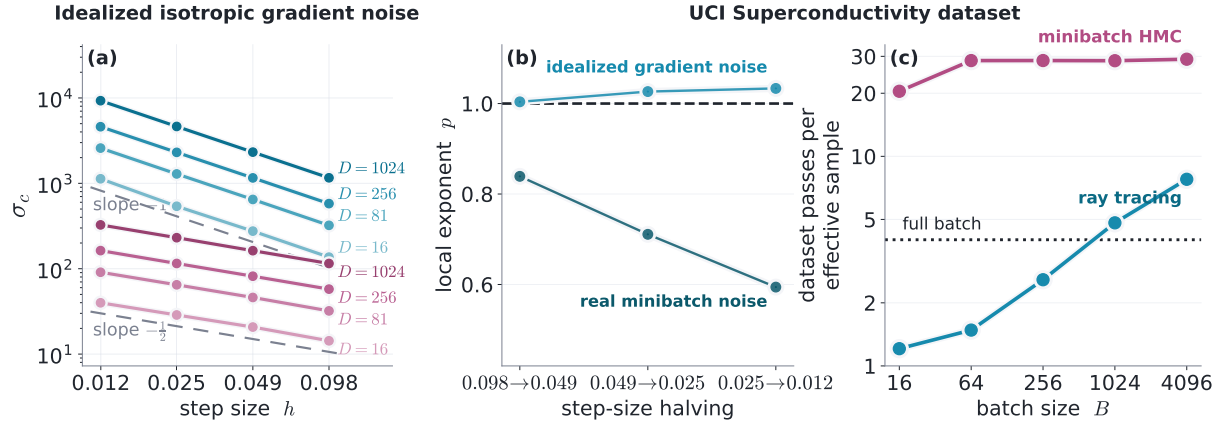


Figure 4 The noise-model ladder. Held-out z standard deviation by rung: 6.88, 6.23, 0.95, 1.08. The target is 1.

level. It samples $\nu = 5.48$, reaches RMSE 0.0480, and cuts the tail count from 108 to 14 against a chance expectation of 1.6 under the fitted t 's own yardstick (Figure 5). It has the best held-out NLL of any method tried, -1.973 .

Convergence and exactness.. Convergence was a campaign in its own right. The batch-1024 arm marched indefinitely. The batch-4096 arm halved σ_{sto} , then decelerated, plateaued, and reversed, with all drift series level at 16M total steps. A drift rule read without the acceptance rate next to it is not evidence of anything.

The exactness check restarted an exact full-batch chain from the converged endpoint. A step-size ladder rejected everything until $dt = 2 \times 10^{-6}$, which accepted at 1.00; production ran 5,000 trajectories at $L = 30$ with acceptance 0.98 and RMSE 0.0469. That stamp certifies the location of the chain, not the width of its spread.

Baselines on the same target.. Table 1 compares methods on the identical heteroscedastic target.

RMSE is close to a tie across all five. Calibration is not. The maximum deviation of any bin from 1 is 0.07 for the chain, 0.21 for the ensemble, and 0.26 for MAP (Figure 6). Only the sampled posterior distributes its uncertainty correctly star by star, which is the property that matters if the

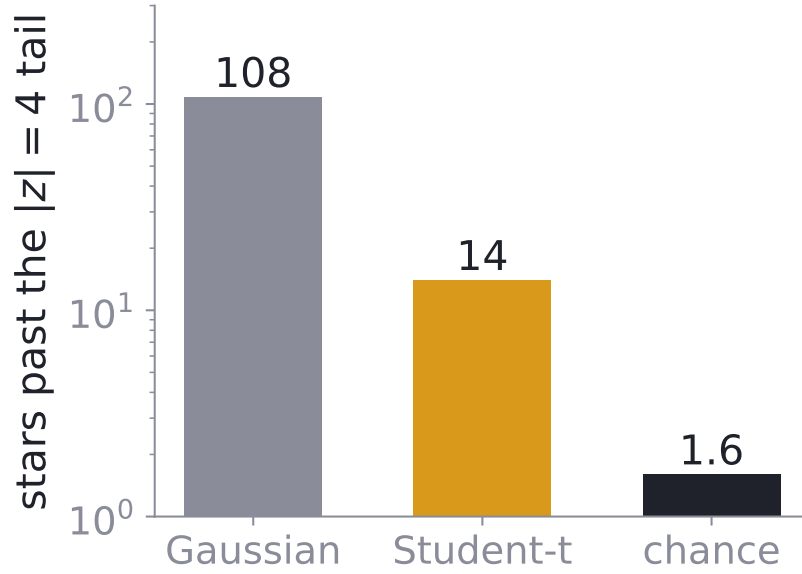


Figure 5 Tail behaviour of the standardized residuals. Moving from a Gaussian to a sampled Student- t likelihood takes the $|z| > 4$ count from 108 to 14.

Table 1 Held-out performance on the heteroscedastic Gaia target (25,232 test stars). Bins are z standard deviation by label error quartile; the target is 1 in every bin.

Method	RMSE	NLL	SD(z)	bins	$ z > 4$
MAP (Adam)	0.0489	-1.733	1.180	1.26 / 1.24 / 1.13 / 1.04	209
Ensemble (10)	0.0473	-1.933	0.962	0.79 / 0.91 / 0.97 / 1.05	61
SGHMC (tuned)	0.0475	-1.875	1.155	1.06 / 1.18 / 1.13 / 1.08	170
RT (Gaussian)	0.0494	-1.941	1.08	0.97 / 1.07 / 1.07 / 1.07	108
RT (Student- t)	0.0480	-1.973	1.12	—	14

error bars are going to be used. MAP’s calibration also anti-correlates with training loss across learning rates: pushing deeper into the mode makes the fit better and the error bars worse.

What the posterior buys.. Averaged over the 25,232 test stars, the mean held-out log predictive density is -1.9731 for the Student- t chain marginal (a log-sum-exp mixture over 50 members) against -1.7332 for the MAP plug-in. The difference is 0.2399 nats, and $e^{0.2399} = 1.271$: held-out stars are on average 27% more probable under the posterior average than under the single best fit. Of that, 0.208 nats (23%) comes from marginalization alone, since the Gaussian chain marginal already reaches -1.9409 , and about 3% comes from the Student- t tails. About two thirds of stars improve individually, with a median improvement near 21%. RMSE is a tie, 0.0489 for MAP against 0.0494 for the chain. The gain is entirely in the error bars, and MAP is overconfident with a z standard deviation of 1.18 .

Against SGHMC.. SGHMC [5] has to tune friction against the gradient-noise scale, and Figure 8 shows what happens when that tuning is off. At friction 30 the kinetic temperature rises from 1.38 to 9.62 across the step sizes tried. At friction 3000 it holds near 1, but is still 8 to 9% hot even when hand-tuned, and its weighted norm was still marching at 2M steps. Ray tracing holds temperature by construction and has no noise-matched knob to get wrong. Two caveats: 2M steps is not 16M, and this is one target. Note also that the heated arm reaches RMSE 0.0474 while

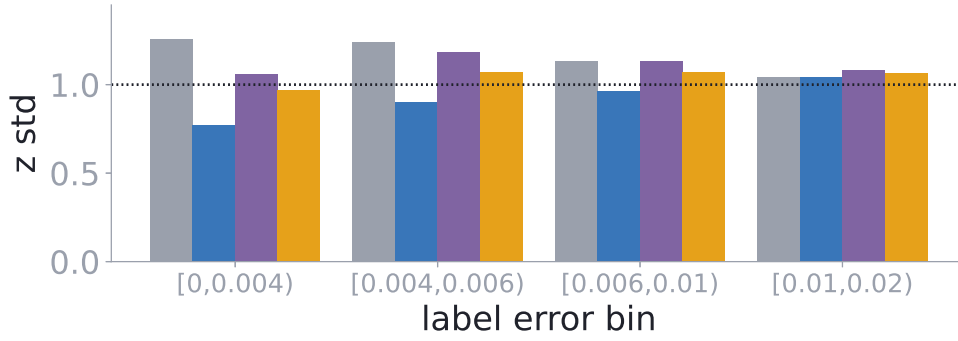


Figure 6 Standardized residual spread by label-error bin for MAP, ensemble, SGHMC, and ray tracing. Flat at 1 is correct.

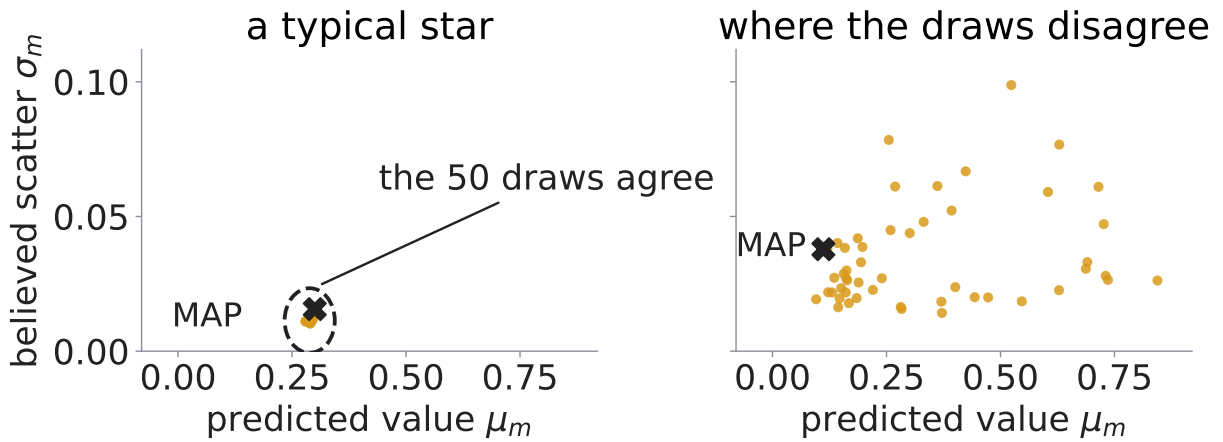


Figure 7 Fifty posterior draws for a typical star and for a star where the draws disagree. The spread is the prediction, not an error bar added afterwards.

boiling. Point accuracy is not evidence of a correct posterior.

Out-of-distribution behaviour.. Using the chain’s τ alarm as an OOD score gives AUC 0.932 on dwarfs, 0.998 on hot stars, and 0.588 on survey-flagged stars, the last of which is near chance and looks like a label-side problem rather than a sampler problem. This should not be read as a ray tracing advantage over SGHMC: the two are statistically indistinguishable here. What OOD detection separates is methods with genuine spread from methods whose spread has collapsed. The contribution of the exactness certificate is the evidence that this spread is genuine.

As a physical sanity check, high- α giants are 20 to 30% brighter below 450 nm at fixed T_{eff} and $[M/H]$, and the network’s calibrated saliency concentrates in the same blue region.

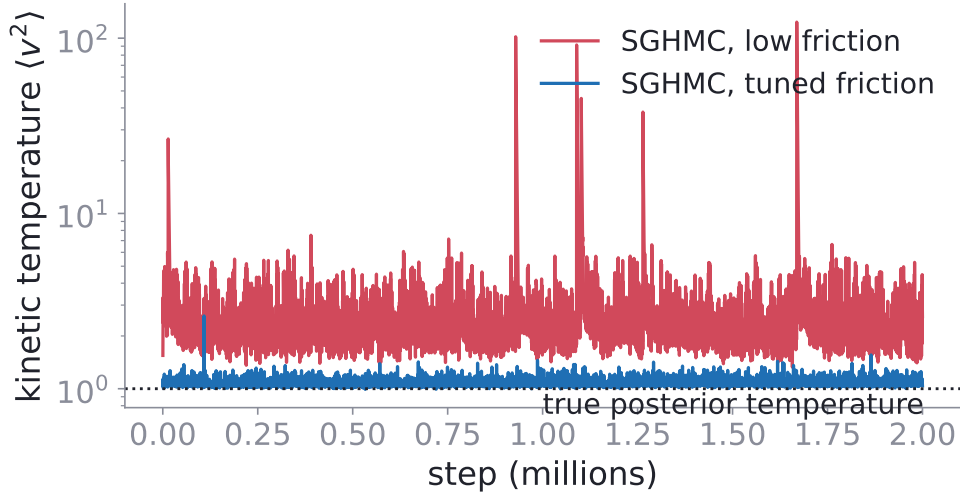


Figure 8 Kinetic temperature for SGHMC at two friction settings against step size, with ray tracing for reference. The friction has to be matched to the noise scale; the ray has no equivalent knob.

4 Why the tolerance differs: the geometry of the auxiliary variable

The measurements above say that the two samplers fail at very different noise levels and with different step-size laws; they do not say why. The why is the subject of a companion manuscript by my mentor [6], written in August 2026 alongside this project. The derivations are his, and I summarize them here without proofs. What this section reports is the part that belongs to the summer: the theory made quantitative predictions, and the measurements were there to meet them.

Two exposures to one noise source.. Minibatching supplies a noisy estimate of the log-density, and a sampler can be exposed to that noise in two places. Differentiating the noisy log-density gives score noise, which perturbs the dynamics of an unadjusted sampler; differencing it between two states gives acceptance noise, which perturbs a Metropolis test. Acceptance noise admits an exact penalty correction [7, 8], but the cost of that correction grows exponentially in the noise level. Score noise has no correction and costs nothing extra. An unadjusted sampler takes the cheap exposure and pays for it with a bias in its stationary distribution; the analysis computes that bias.

One identity for the stationary bias.. For a sampler that alternates deterministic transport with a refresh, the stationary bias is injection over relaxation: $b = J/(1 - \lambda)$, the noise injected per step divided by the fraction the refresh removes. The identity recovers the classical results. Unadjusted Langevin gives $b = \sigma^2 h/2$ to leading order, the known SGLD bias. Unadjusted HMC gives $b = c(S) \sigma^2 h$ with $c(S) = (2S - \sin 2S)/(4 \sin^2 S)$, which diverges at the half-orbit $S = \pi$: the refresh stops contracting exactly where the noise keeps arriving. Minibatch HMC does not just carry a bias, it carries a resonance.

The ray tracing result.. For ray tracing the same identity gives $b = \sigma^2 h^2/[12(D - 1)]$: independent of trajectory length, one order higher in the step size than HMC, and with no resonance at any length, because the noise forces exactly the mode that the refresh relaxes, so injection and relaxation vanish together where the resonance would sit. The mechanism in words: the direction lives on a compact sphere whose equilibrium law is uniform, and isotropic noise acts as a random rotation, which leaves a uniform law invariant, so the leading effect of the noise cancels. What survives is the failure of the noise to commute with the deterministic bending inside a single step,

and that is second order in h .

Anisotropy, the caveat that matches the measurement.. Real minibatch noise is not isotropic, and the h^2 law is the isotropic result. With noise covariance C , the sampled covariance picks up a distortion of $\sigma^2 h S(C - I)/(3D)$: first order in h , so a smaller step cannot refine it away; growing with trajectory length; and traceless, so mean-squared-radius diagnostics never see it. This is the term behind the measurement in Section 3.2, where ray tracing’s local step-size exponent weakened from about 1 toward 0.6 once idealized isotropic noise was replaced by real minibatch noise. The practical advice follows directly: estimate the noise covariance and whiten it, and the h^2 law is bought back.

Confrontation with measurement.. The predicted step-size exponents ($\sigma_c \propto h^{-1/2}$ for HMC, $\sigma_c \propto h^{-1}$ for ray tracing) match both the slopes measured here under idealized noise (Section 3.2) and the slopes in Behroozi’s $D = 10,000$ experiment [1]. The predicted absolute tolerances, with no fitted quantities, land within 30% of that experiment: theory $\sigma_c^2 = 2.67$ against a reported 3.2 for ray tracing; 0.0085 against 0.012 for HMC; a predicted variance ratio of 314 against the reported ratio above 250. For a first-principles calculation with nothing tuned, that is close.

Batch economics.. Since $\sigma^2 \propto 1/B$, holding accuracy fixed forces $h \propto B^{-1/2}$ for HMC, and total compute is then flat in B : shrinking the batch buys nothing. Ray tracing needs only $h \propto B^{-1/4}$, so total compute scales as $\sqrt{B}$, and a hundredfold smaller batch costs ten times less. This is the theoretical version of the cost frontier measured in Section 3.2.

Two design warnings.. The analysis also warns against two plausible modifications. Splitting the noise out of the kick, applied instead as an independent rotation of matched variance, triples the bias; and at matched gradient cost the minimal-norm integrator carries about 20% more noise bias than plain leapfrog, despite its smaller clean discretization error. Matching the variance of a noise source is not the same as matching its effect. On the constructive side, the analysis supports Behroozi’s conjecture that a constant-speed variant of HMC would behave like ray tracing: the proofs use only the compactness of the auxiliary domain and the rotational invariance of its equilibrium law, and nothing in them refers to the optical construction.

What exact adjustment changes.. The second companion manuscript, Speagle [9], asks what the advantage is worth once the Metropolis test itself is corrected for minibatch noise, and the answer sharpens every claim in this report. With an exact minibatch adjustment in place, ray tracing and HMC tolerate the same noise variance to within 0.05% at matched decorrelation and gradient budget: the factor above 250 is a property of *unadjusted* sampling, not of the dynamics as such. What the adjustment removes differs sharply between the two samplers. At the operating point, ray tracing’s unadjusted bias is smaller than HMC’s by a factor that grows with trajectory length, about $6T$, so adjustment takes almost nothing away from ray tracing; per squared step size the residual bias is $1/[12(D - 1)T]$ for ray tracing against $1/(2D)$ for HMC. A lower bound on the product of batch size and trajectory length, $BT \geq c \Lambda^2/s_*^2$, adds the economic half: minibatching and exact adjustment are alternatives, not complements, since paying for one removes the reason to pay for the other. The honest summary of the whole report follows. The large unadjusted advantage is real, but it is an advantage of unadjusted sampling; adjusted, the two samplers reach tolerance parity; ray tracing is the sampler for which skipping the adjustment is nearly free.

5 Discussion

Three things came out of the summer that are worth stating separately from the numbers.

The geometry argument holds, and it is now quantitative.. At the start of the project the claim was a picture: a bounded direction cannot heat the way an unbounded momentum can. That picture is now attached to measured exponents (h^{-1} against $h^{-1/2}$), to a closed-form bias ($\sigma^2 h^2 / [12(D-1)]$), and to agreement between the two within 30% on an experiment neither was fitted to. It also survived contact with a real problem: the Gaia chain converged under minibatch gradients at $D = 11,394$ and then passed an exact full-batch test from its endpoint.

The sampler was rarely the hard part.. On the Gaia campaign the largest single improvement, by a wide margin, came from fixing the noise model, not from anything about sampling: z standard deviation 6.88 to 1.08. The second largest came from where the chains started. The third came from reading drift diagnostics correctly, which cost weeks of wall-clock on arms that were marching rather than converged. A sampler with better noise tolerance removes one obstacle. It does not remove the modelling work, and on this problem the modelling work was larger.

The payoff shows up in the error bars, not the predictions.. Across five methods on the same Gaia target, RMSE spans 0.0473 to 0.0494, which is close to a tie. Calibration spans a maximum bin deviation of 0.07 to 0.26, which is not. The 27% mean improvement in held-out predictive density is almost entirely marginalization: 23 of the 27 points come from averaging over the posterior, and only about 3 from the heavier tails. This is the practical case for paying for a sampler, and it is also the reason accuracy comparisons are a poor way to evaluate one. The SGHMC arm that was visibly boiling still reached RMSE 0.0474.

On comparison claims.. Two comparisons that would have been convenient to make are not supported by the evidence collected here. The out-of-distribution results do not separate ray tracing from SGHMC in any statistically meaningful way; what they separate is genuine spread from collapsed spread, and both samplers have genuine spread. And the speed comparison is weaker than the noise-tolerance comparison: at matched accuracy the gate-passing saving was about $1.5\times$ at batch 256 on one problem. The published variance ratio above 250 at $D = 10,000$ [1] is a noise-tolerance number, not a wall-clock speedup, and it belongs to that paper rather than to this work.

6 Limitations and future work

It is worth being explicit about which claims here are settled and which are not.

Validated.. Robustness to stochastic-gradient noise. This has a mechanism (bounded auxiliary variable on a compact sphere), measured scalings on both idealized and real noise, an independent theoretical derivation that agrees with the published numbers to within 30%, and a working application on a real survey dataset at $D = 11,394$ that passed an exact full-batch check.

Partial.. The cost story. The honest statement is that at matched accuracy the gate-passing saving was about $1.5\times$ at batch 256, on one problem. The measured cost frontier used an assumed full-batch reference rather than a measured one, and the theory's prediction of $O(\sqrt{B})$ economics has not been tested end to end on a real target. Any larger speedup figure from these runs comes from configurations that fail their own accuracy gate and should not be quoted.

Open..

- **Width honesty.** The Student- t chain still shows about 4 to 5% PIT under-coverage. The exact finisher certifies where the chain sits, not how wide it is, so this is currently unresolved by the diagnostics available.

- **Anisotropy.** The theory says the surviving first-order bias term is traceless and that whitening the noise covariance should recover the h^2 law. Neither the whitening nor a direct measurement of that traceless distortion was attempted here. This is the most concrete next experiment.
- **Single chains.** Every production result comes from one chain per configuration. There is no multi-chain $\hat{R}$, so the convergence evidence is drift-based and within-chain.
- **Scale.** $D = 11,394$ is a real network but it is far below the scale where Bayesian deep learning is usually argued about [10, 11]. Whether the h^2 law survives at that scale with real anisotropic noise is untested.
- **Mode coverage.** Nothing in this project validates behaviour on multimodal or disconnected targets. The published method has a separate emission rule for that case; it was not exercised here and no claim is made about it.
- **Learning the weighting function.** The sampler's family is parameterized by a weighting function W that recovers HMC, microcanonical HMC, Gibbs, and tempering as special cases, and fair sampling holds for any choice. Learning W for a given target, suggested by Prof. Ricardo Baptista, is the natural research direction out of this work and was not attempted.

One disclosure.. The rung-1 production gate ran roughly $49\times$ too soft, because σ_{sto} had been measured at a warm start that a bug had scrambled. This was caught on 2026-07-30. The validity of the results reported here rests on the exact full-batch finisher and the drift checks, which are unaffected, and the chain of record is the rung-2 heteroscedastic chain rather than rung 1.

7 Conclusion

The question this project set out to answer was whether replacing an unbounded momentum with a bounded direction makes minibatch MCMC practical. On the evidence collected here the answer is yes for the noise problem specifically, and the size of the effect is now known rather than argued.

Three results carry that conclusion. The critical noise level scales as h^{-1} for ray tracing against $h^{-1/2}$ for HMC, measured directly. The companion analysis derives a stationary bias of $\sigma^2 h^2 / [12(D - 1)]$, one order higher in h , with no resonance at any trajectory length, and it reproduces the published tolerances to within 30% with nothing fitted. And on a real target, 126,156 Gaia giants and an 11,394 parameter network, a minibatch chain converged, passed an exact full-batch check from its endpoint, and produced held-out predictions 27% more probable on average than the single best fit, with flat calibration across every label-error bin where MAP, a deep ensemble, and SGHMC are all visibly off.

The rest of the summer was a reminder of where the difficulty actually sits. Fixing the noise model moved the calibration by a factor of six; the sampler moved it by a fraction of that. Diagnostics had to be read in pairs, because a drift series without an acceptance rate beside it says nothing. And point accuracy turned out to be almost useless as a signal, since a chain that was demonstrably overheating still matched everyone else's RMSE.

What remains open is the width of the posterior, the effect of anisotropic noise in practice, and whether any of this holds at the scale of a modern network. The most direct next step is the one the theory points at: estimate the minibatch noise covariance, whiten it, and check whether the second-order law comes back.

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